The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 August 2011.

INCH–POUND

MIL–PRF–19500/574C
27 May 2011
SUPERSEDING
MIL–PRF–19500/574B
30 October 1998

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, LIGHT EMITTING,
TYPES 1N6497, 1N6498, 1N6499, 1N6503, 1N6504, AND 1N6505,
JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL–PRF–19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for red, yellow, and green light-emitting diodes (LED) with internal current regulation requiring no external resistors for operation on any voltage from 3 V dc to 30 V dc. Two levels of product assurance are provided for each device type as specified in MIL–PRF–19500.

1.2 Physical dimensions. See figure 1 for 1N6497 through 1N6499 and figure 2 for 1N6503 through 1N6505.

1.3 Maximum ratings. Unless otherwise specified T_A = +25°C.

V_F	$V_{(BR)}$ I_R = 10 μA dc	P_{FM} (1)	T_{op} and T_{stg}
V dc	V dc	mW(pk)	°C
30	5	225	-65 to +100

(1) Derate at 3.0 mW/°C above +25°C.

1.4 Characteristics, radiometric (physical), and photometric (visual).

Type	Color	V_F	I_V Θ = 0°	I_F		λV		I_R at V_R = 3 V	C at V_R = 0 V
		V dc	mcd	mA dc		nm		μA dc	pF
			Min	Min	Max	Min	Max		
1N6497, 1N6503	Red	20	.5	3.5	7.5	595	695	1	500
1N6498, 1N6504	Yellow	20	.5	3.5	7.5	570	595	1	500
1N6499, 1N6505	Green	20	.5	3.5	7.5	525	580	1	500

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL–PRF–19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL–STD–750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

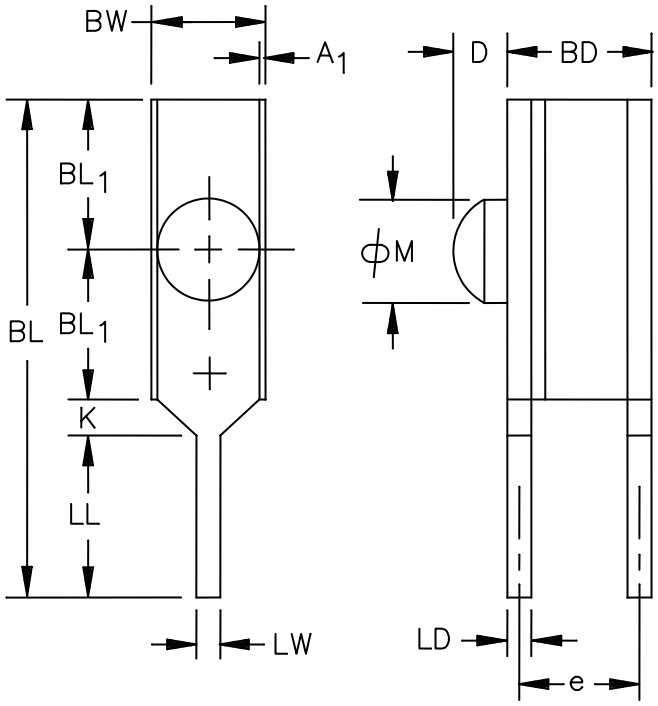
3.1 General. The individual item requirements shall be as specified in MIL–PRF–19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL–PRF–19500 and as follows.

I_{FM}	Forward current (subscript M indicates maximum).
I_V	Luminous intensity (the subscript V is used to designate a photometric or visual quantity to differentiate from I used herein for current).
λ_V	Peak radiometric wavelength of diode light emission.
mcd	Milli-candela; the candela is a unit of luminous intensity defined such that the luminance of a blackbody radiator at the temperature of solidification of platinum is 60 candelas per square centimeter.
P_{FM}	Forward power dissipation (the subscript M indicates maximum).
Θ	The angle at or off the axis of symmetry of a light source at which luminous intensity is measured.

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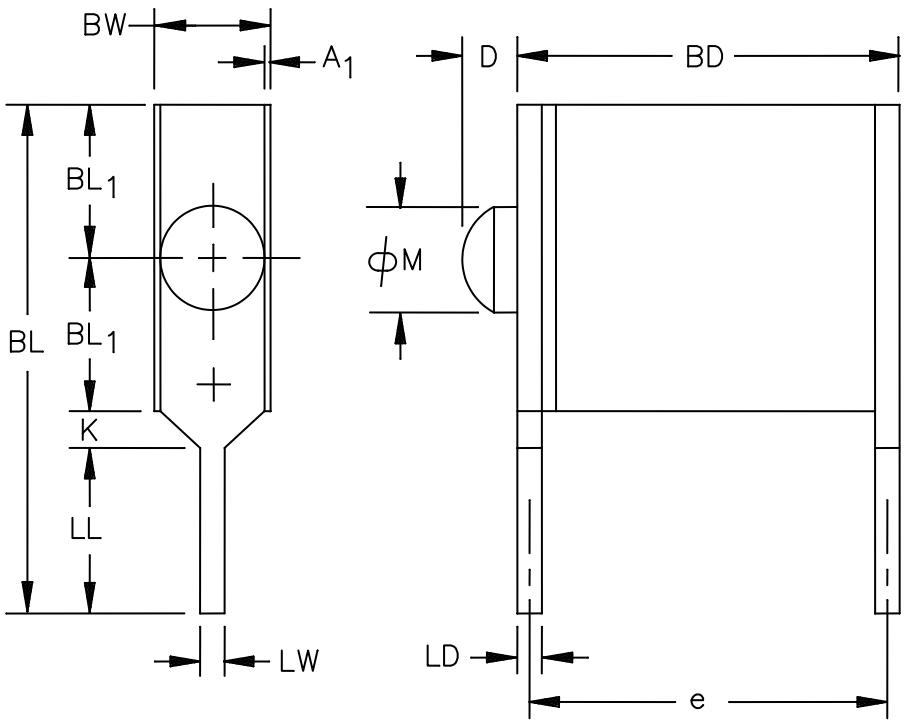


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A_1	.003	.005	0.08	0.13	2
BD	.112	.128	2.84	3.25	
BL	.385	.445	9.78	11.30	
BL_1	.123	.127	3.12	3.23	
BW	.093	.099	2.36	2.51	
D	.045 Nominal		1.14 Nominal		
e	.100 BSC		2.54 BSC		3
K	.015	.045	0.38	1.14	
LD	.018	.022	0.46	0.56	
LL	.125	.145	3.17	3.68	
LW	.020	.022	0.51	0.56	
∅M	.075	.082	1.90	2.08	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The front and back pins are recessed on the two sides to prevent shorting of an adjacent device.
3. The basic pin spacing is between centerlines.
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions for types 1N6497, 1N6498, and 1N6499.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A_1	.003	.005	0.08	0.13	2
BD	.312	.328	7.92	8.33	
BL	.385	.445	9.78	11.30	
BL_1	.123	.127	3.12	3.23	
BW	.093	.099	2.36	2.51	
D	.045 Nominal		1.14 Nominal		
e	.300 BSC		7.62 BSC		3
K	.015	.045	0.38	1.14	
LD	.018	.022	0.46	0.56	
LL	.125	.145	3.17	3.68	
LW	.020	.022	0.51	0.56	
∅M	.075	.082	1.90	2.08	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The front and back pins are recessed on the two sides to prevent shorting of an adjacent device.
3. The basic pin spacing is between centerlines.
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 2. Physical dimensions for types 1N6503, 1N6504, and 1N6505.

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3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL–PRF–19500, and on figures 1 and 2.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL–PRF–19500, MIL–STD–750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL–PRF–19500. Manufacturer's identification and date code shall be marked on the devices. Initial container package marking shall be in accordance with MIL–PRF–19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL–PRF–19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANTX only). Screening shall be in accordance with table E–IV of MIL–PRF–19500 and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (See table E-IV of MIL–PRF–19500)	Measurements
	JANTX level
7	Test method 1071 of MIL–STD–750, fine leak, test condition H (leak testing 30 minutes after pressurization is acceptable). Test method 1071 of MIL–STD–750, gross leak, test condition C, except that leak indicator fluid shall be maintained at +100°C ±5°C.
9 and 10	Not applicable.
11	Subgroup 2 of table I herein.
12	V_F = 30 V dc; T_A = +25°C, t = 96 hours.
13	Subgroup 2 of table I herein; ΔI_{V1} = -20 percent of initial readings. ΔI_F = ±1 mA dc.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL–PRF–19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E–V of MIL–PRF–19500, table I herein, and as specified herein. Electrical measurements (end-points) shall be in accordance with subgroup 2 of table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E–VIb of MIL–PRF–19500 and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with subgroup 2 of table I herein.

4.4.2.1 Group B inspection, appendix E, table E–VIb of MIL–PRF–19500.

Subgroup	Method	Condition
B2	1051	Test condition A, except T(high) = +100°C (10 cycles); time at temperature extremes 15 minutes minimum.
B2	1071	Fine leak: Test condition H (leak testing 30 minutes after pressurization is acceptable). Gross leak: Test condition C, except that leak indicator fluid shall be maintained at +100°C ±5°C.
B3	1027	V_F = 30 V dc; T_A = +25°C; t = 340 hours.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E–VII of MIL–PRF–19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of subgroup 2 of table I herein.

4.4.3.1 Group C inspection, appendix E, table E–VII of MIL–PRF–19500.

Subgroup	Method	Condition
C2	1056	Test condition A.
C2	2036	Test condition E.
C2	1071	Fine leak: Test condition H (leak testing 30 minutes after pressurization is acceptable). Gross leak: Test condition C, except that leak indicator fluid shall be maintained at +100°C ±5°C.
C3	2016	Nonoperating; 1,500 G's; t = 0.5 ms; 5 blows in each orientation: X1, Y1, and Y2.
C3	2056	Nonoperating.
C3	2006	Nonoperating; 20,000 G's; X1, Y1, and Y2.
C6	1026	V_F = 30 V dc; T_A = +25°C.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Luminous intensity. This measurement is made with a calibrated photometer.

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TABLE I. Group A inspection.

Inspection 1/	MIL–STD–750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1	2071					
Visual and mechanical examination						
Subgroup 2						
Luminous intensity		Θ = 0 degrees (see 4.5.1), V_F = 5 V dc	I_V			mcd
1N6497, 1N6498				0.5		
1N6503, 1N6504				0.5		
1N6499, 1N6505				0.5		
Luminous intensity		Θ = 15 degrees (see 4.5.1), V_F = 5 V dc	I_{V2}			mcd
1N6497, 1N6498				0.25		
1N6503, 1N6504				0.25		
1N6499, 1N6505				0.25		
Reverse current	4016	DC method; V_R = 3 V dc	I_R		1.0	μA dc
Forward voltage	4026	DC method	I_F			
1N6497,1N6498,1N6499 1N6503,1N6504,1N6505		V_F = 30 V dc		3.5	7.5	mA dc
1N6497,1N6498,1N6499 1N6503,1N6504,1N6505		V_F = 5 V dc		3.5	7.5	mA dc
Subgroup 3						
High temperature:		T_A = +100°C				
Reverse current	4016	DC method; V_R = 3 V dc	I_R		1.0	μA dc
Forward voltage	4026	DC method	I_F	3.0		mA dc
1N6497,1N6498,1N6499 1N6503,1N6504,1N6505		V_F = 30 V dc				
1N6497,1N6498,1N6499 1N6503,1N6504,1N6505		V_F = 5 V dc				

See footnote at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL–STD–750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 – Continued						
Low temperature:		T_A = -55°C				
Reverse current	4016	DC method; V_R = 3 V dc	I_R		1.0	μA dc
Forward current	4026	DC method	I_F		15.0	mA dc
1N6497,1N6498,1N6499 1N6503,1N6504,1N6505		V_F = 30 V dc				
1N6497,1N6498,1N6499 1N6503,1N6504,1N6505		V_F = 5 V dc				
Subgroup 4						
Capacitance	4001	V_R = 0; f = 1 MHz	C		500	pF
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plans, see MIL–PRF–19500.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL–STD–750		Qualification inspection
	Method	Conditions	
Subgroup 1			n = 45, c = 0
Thermal shock (glass strain)	1056	100 cycles 0°C to 100°C	
Temperature cycling	1051	500 cycles, -65°C to +100°C	
Hermetic seal	1071		
Electrical measurement		See subgroup 2 of table I herein.	
Subgroup 2			
Intermittent operating life	1037	10,000 cycles.	
Electrical measurements		See subgroup 2 of table I herein.	
Subgroups 4 and 5			
Not applicable			
Subgroup 6			n = 11, c = 0
ESD	1020		
Subgroups 7, 8, and 9			
Not applicable			

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL–PRF–19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil .

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force – 85
DLA – CC

Review activity:
Air Force – 99

Preparing activity:
DLA – CC

(Project 5980–2006–010)

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